

Slide 1 – Title: “Ecosystem Analytics – Using Data to Optimize Collaboration and Growth”

■■■ Say: “This presentation is about Ecosystem Analytics — the use of data to understand and improve how different parts of a software ecosystem work together. We'll see how analytics helps in collaboration, innovation, and long-term software growth.”

Slide 2 – Introduction

■■■ Say: “A software ecosystem is just like a biological one — many interconnected entities like projects, developers, organizations, and users working together. Ecosystem Analytics helps us understand how these entities interact and evolve using data. The main goal is to make the ecosystem more efficient and healthy.”

Slide 3 – Importance of Software Ecosystem Analytics

■■■ Say: “This field is important because it helps us identify who contributes most, which modules are popular, and where improvements are needed. It improves decision-making, detects inactive or weak areas, and encourages a data-driven development culture.”

Slide 4 – Data Sources in Ecosystem Analytics

■■■ Say: “To perform ecosystem analytics, we collect data from several sources like:

Code Repositories such as GitHub and GitLab (commits, pull requests, issues)

Package Dependencies like npm, PyPI, or Maven

Community Engagement metrics such as stars, forks, and downloads

Usage Metrics like app adoption and bug reports.”

Slide 5 – Key Metrics for Analysis

■■■ Say: “These are some major metrics we analyze:

Developer Activity: how often contributors make changes

Project Health: how quickly issues are resolved or updates are released

Dependency Centrality: how critical a module is to others

Collaboration Intensity: how frequently developers work together.”

Slide 6 – Optimization in Software Ecosystems

■■■ Say: “Analytics helps optimize different areas:

Collaboration – balancing workload among contributors

Innovation – spotting new trends early

Quality – finding fragile or outdated dependencies. Overall, this leads to continuous improvement and sustainable growth.”

Slide 7 – Case Study: GitHub Ecosystem

■■■ Say: “GitHub is a great example of how ecosystem analytics works. By studying metrics like active repositories, pull requests, and trending projects, we can understand developer behavior and predict which projects will succeed.”

Slide 8 – Case Study: Android Ecosystem

■■■ Say: “Google uses ecosystem analytics for Android. It gathers data from millions of apps to maintain compatibility and performance. Companies like Palantir help integrate this data. This approach helps Google identify device fragmentation and prioritize software updates.”

Slide 9 – Benefits of Ecosystem Analytics

■■■ Say: “Ecosystem Analytics provides many benefits:

Encourages innovation through data-driven insights

Improves collaboration and trust

Reduces maintenance costs

Enables better long-term planning for software evolution.”

Slide 10 – Challenges and Limitations

■■■ Say: “However, there are challenges:

Data privacy issues with sensitive user or contributor data

Complex analysis due to large datasets

Dynamic nature since ecosystems change rapidly

Bias in metrics, as not all data reflects true collaboration.”

Slide 11 – Future of Software Ecosystem Analytics

■■■ Say: “In the future, we’ll see:

AI/ML models predicting ecosystem growth

Visualization tools for real-time monitoring

Predictive maintenance for code dependencies

A shift toward fully data-driven governance.”

Slide 12 – Conclusion

■■■ Say: “To sum up — software ecosystems rely on collaboration and innovation. Ecosystem Analytics converts raw data into useful insights, helping software grow sustainably and efficiently.”

Slide 13 – Thank You

■■■ Say: “Thank you! I hope this helped you understand how data and analytics play a major role in software ecosystems.”